

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0733

Bluebonnet 230 kV
September 26, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0733. The Interconnection Customer proposes a 438.6 MW solar photovoltaic facility interconnecting at the Bluebonnet 230 kV substation in the Indiana area. The request was received on January 21, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 265 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,684,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Calwell-Blackrock 345 kV	Carbon Ctr 500/345 kV Auto	101.9	1,217	5.5
Piney Crk Project 500/345kV Transformer	Hilltop-Piney Creek Project 500 kV	100.8	1,923	8.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Calwell-Blackrock 345 kV	Reconductor line	\$19,997,000	4% (\$766,000)
Piney Crk Project 500-345 kV Transformer	Replace transformer with higher-rated unit	\$23,976,000	4% (\$918,000)
	Total allocated to project		\$1,684,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.